Visualizing US Natural Disaster Declarations: Trends and Patterns

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MILESTONE 1

Data Acquisition, Cleaning & Initial Exploration

Objective

The objective of Milestone-1 is to prepare the FEMA disaster dataset for analysis by:

- Loading the dataset
- Cleaning missing and inconsistent values
- Validating data types (especially dates)
- Performing initial exploratory data analysis (EDA)
- Creating preliminary visualizations

This milestone ensures the dataset is reliable before performing advanced analysis.

Data Acquisition & Loading

Tasks Performed:

- Imported required libraries (Pandas, NumPy, Matplotlib)
- Loaded FEMA dataset into DataFrame
- Checked dataset shape (rows × columns)
- Displayed first 5 records
- Verified column names

```
# STEP 1: Import Libraries
```

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# show plots inside notebook
%matplotlib inline
```

```
# STEP 2: Load the Dataset
```

```
file_path = r"C:\Users\Admin\Downloads\database.csv.zip"

df = pd.read_csv(file_path, compression='zip')

# check first rows
df.head()
```

	Declaration Number	Declaration Type	Declaration Date	State	County	Disaster Type	Disaster Title	Start Date	End Date	Close Date	Individual Assistance Program	Individuals & Households Program	Public Assistance Program	Hazard Mitigation Program
0	DR-1	Disaster	05/02/1953	GA	NaN	Tornado	Tornado	05/02/1953	05/02/1953	06/01/1954	Yes	No	Yes	Yes
1	DR-2	Disaster	05/15/1953	TX	NaN	Tornado	Tornado and Heavy Rainfall	05/15/1953	05/15/1953	01/01/1958	Yes	No	Yes	Yes
2	DR-3	Disaster	05/29/1953	LA	NaN	Flood	Flood	05/29/1953	05/29/1953	02/01/1960	Yes	No	Yes	Yes
3	DR-4	Disaster	06/02/1953	MI	NaN	Tornado	Tornado	06/02/1953	06/02/1953	02/01/1956	Yes	No	Yes	Yes
4	DR-5	Disaster	06/06/1953	MT	NaN	Flood	Floods	06/06/1953	06/06/1953	12/01/1955	Yes	No	Yes	Yes

Outcome:

- Dataset successfully loaded
- Identified total records and available features
- Observed important columns like:
  - declarationDate
  - stateName
  - incidentType
  - ihProgramDeclared
  - paProgramDeclared

Data Cleaning

Tasks Performed:

- Checked missing values
- Removed duplicate records
- Converted declarationDate to datetime format
- Extracted year and month
- Standardized incidentType values

Initial Exploratory Data Analysis (EDA)

Analysis Performed:

- Total disaster declarations
- Number of unique states
- Number of unique incident types
- Disaster count per state
- Disaster count per year

Visualizations

Visualization 1: Declarations per Year

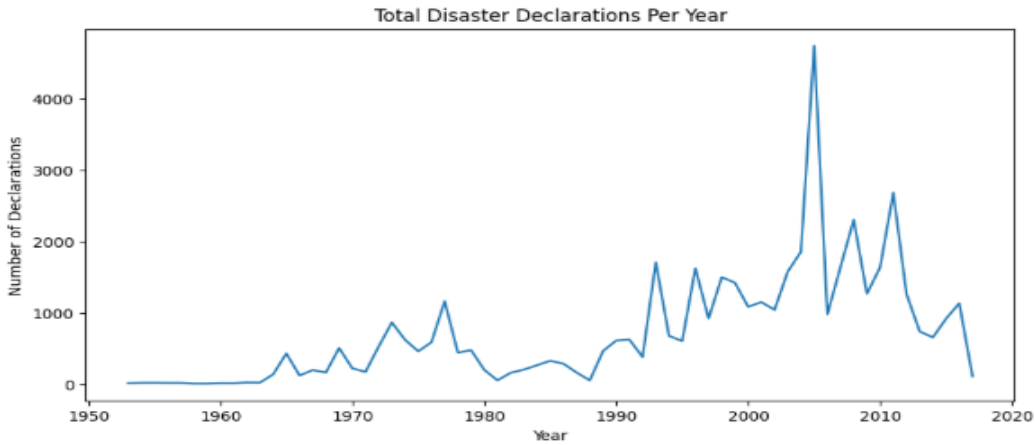
- X-axis → Year
- Y-axis → Count of declarations
- Type → Line chart

```
# 1. Total Declarations Per Year -
```

```
year_data = df.groupby('Year').size()

import matplotlib.pyplot as plt

plt.figure(figsize=(10,5))
plt.plot(year_data.index, year_data.values)
plt.title("Total Disaster Declarations Per Year")
plt.xlabel("Year")
plt.ylabel("Number of Declarations")
plt.show()
```



Visualization 2: Declarations per State

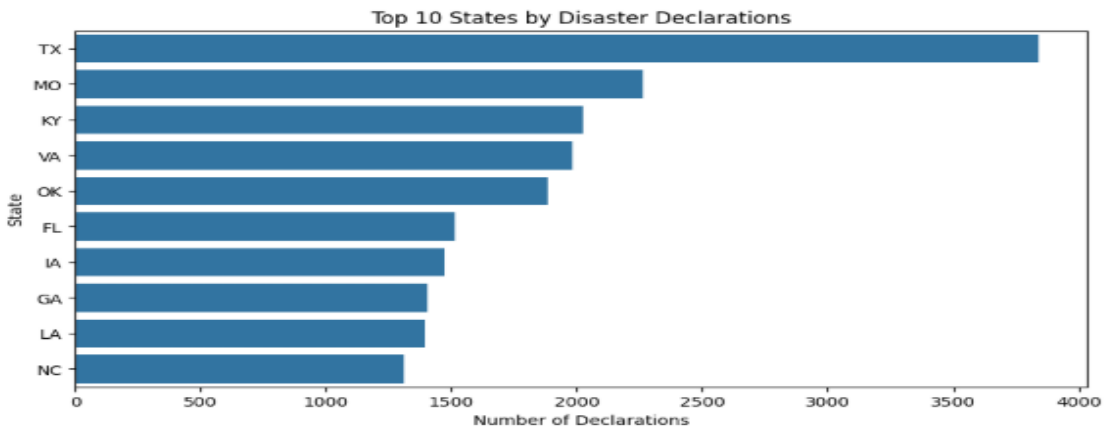
- X-axis → State
- Y-axis → Count
- Type → Bar chart

```
# 2. Time Series Plot
```

```
import seaborn as sns
import matplotlib.pyplot as plt

state_data = df['State'].value_counts().head(10)

plt.figure(figsize=(10,5))
sns.barplot(x=state_data.values, y=state_data.index)
plt.title("Top 10 States by Disaster Declarations")
plt.xlabel("Number of Declarations")
plt.ylabel("State")
plt.show()
```



All plots included:

- Title
- Axis labels
- Proper scaling

Key Insights

- Dataset spans multiple decades of disaster declarations.
- Some states show significantly higher disaster frequency.
- Initial trend indicates an increase in disaster declarations over time.
- Incident types vary widely across regions.

Conclusion

Milestone-1 successfully prepared the dataset for analysis.
All inconsistencies were handled, date formats were validated, and basic trends were identified.

This milestone shows a strong foundation for geographical pattern analysis in the next stages.

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